

Problem-Solution Fit

Date	2 JULY 2026
Team ID	XXXXXX
Project Name	Flood Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Disaster Management Authorities • Meteorologists • Government Agencies • Local Administration • People living in flood-prone areas	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited access to real-time weather data • Lack of advanced prediction tools • Limited technical expertise • Internet connectivity issues in rural areas • Budget constraints for advanced monitoring system	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing Solutions • Traditional weather forecasting • Manual flood monitoring • Government warning systems Pros • Easy to access • Uses official weather information • Supports emergency planning Cons • Limited prediction accuracy • Delayed warnings • Cannot analyze multiple weather parameters efficiently
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Delayed flood warnings during	9. PROBLEM ROOT / CAUSE [RC] ☐ Climate change causing	7. BEHAVIOR + ITS INTENSITY [BE] ☐ Regularly monitor rainfall reports during

heavy rainfall • Inaccurate flood prediction methods • Loss of lives and property due to late response • Frequent flooding during monsoon seasons	unpredictable rainfall ☐ Insufficient early-warning systems ☐ Delayed analysis of weather data ☐ Dependence on traditional forecasting methods ☐ Lack of machine learning-based prediction systems	monsoon ☐ Check weather forecasts frequently ☐ Respond immediately to flood alerts ☐ High urgency during heavy rainfall events
3. TRIGGERS TO ACT [TR] • Heavy rainfall alerts • Rising river or water levels • Extreme weather forecasts • Sudden climate changes • Requests for early disaster preparedness	10. YOUR SOLUTION [SL] Develop a Machine Learning-based Flood Prediction System that analyzes historical weather data, rainfall, cloud visibility, and seasonal patterns using Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and XGBoost algorithms. The best-performing model (XGBoost – 96.55% accuracy) is integrated into a Flask web application and can be deployed on IBM Cloud to provide fast, accurate, and reliable flood risk predictions, helping authorities issue early warnings and improve disaster preparedness.	8. CHANNELS OF BEHAVIOR [CH] ONLINE ☐ Flask Web Application ☐ IBM Cloud Platform ☐ Government Weather Portals ☐ Mobile/Web Browser
4. EMOTIONS BEFORE / AFTER [EM] Before Using the System • Worried • Uncertain • Stressed • Unprepared After Using the System • Confident • Prepared • Relieved • Safe		OFFLINE ☐ Disaster Management Offices ☐ Emergency Control Rooms ☐ Local Government Authorities ☐ Community Awareness Programs